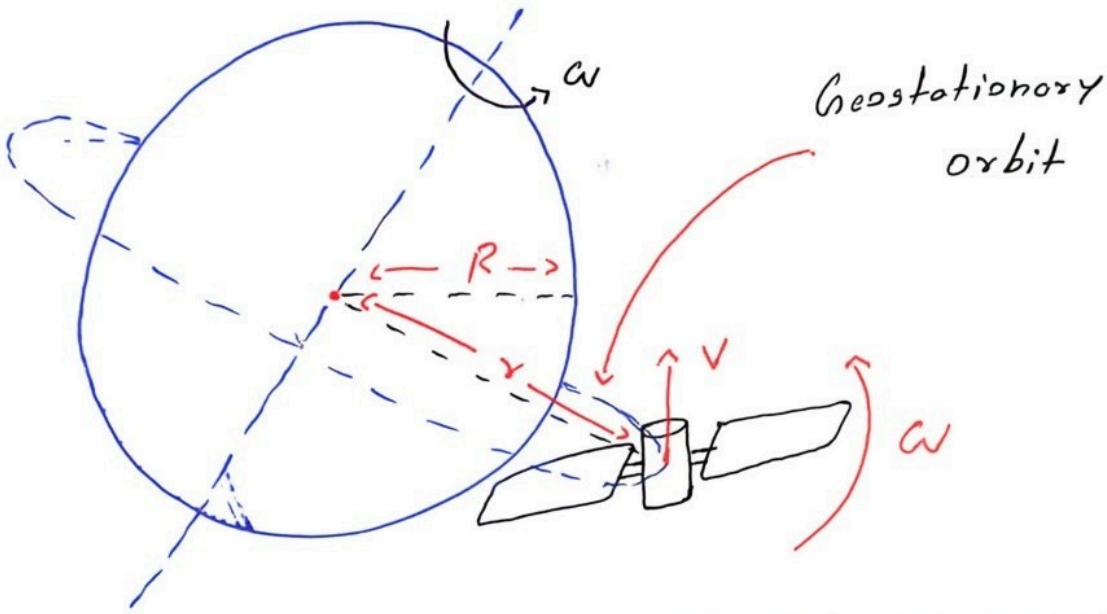
Gravitational Field

⇒ Geostationary Satellite



① Periodic time of geostationary satellite = Periodic time of earth

② Angular velocity of geostationary satellite = Angular velocity of earth

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→ height of the geostationery Orbit.

$$\frac{GMm}{r^2} = mr\omega^2$$

$$\Downarrow$$

$$GM = r^3 \times \left(\frac{2\pi}{T}\right)$$

$$r = \left(\frac{gR^2T^2}{4\pi^2}\right)^{\frac{1}{3}}$$

$$r = \left(\frac{GMT^2}{4\pi^2}\right)^{\frac{1}{3}}$$

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